Since $|A|$ is in cyclic symmetric form in p, q, r and hence q and r also factors.

Putting $p + q + r = 0 \Rightarrow q + r = -p$; $r + p = -q$; and $p + q = -r$.

$$|A| = \begin{vmatrix} p^2 & p^2 & p^2 \\ q^2 & q^2 & q^2 \\ r^2 & r^2 & r^2 \end{vmatrix} = 0$$ since 3 columns are identical.

Therefore, $(p + q + r)^2$ is a factor of $|A|$.

The degree of the obtained factor $pqr\,(p + q + r)^2$ is 5. The degree of $|A|$ is 6.

Therefore, required factor is $k\,(p + q + r)$.

$$\begin{vmatrix} (q+r)^2 & p^2 & p^2 \\ q^2 & (r+p)^2 & q^2 \\ r^2 & r^2 & (p+q)^2 \end{vmatrix} = k(p+q+r)\,(p+q+r)^2 \times pqr$$

Taking $p = 1$, $q = 1$, $c = 1$, we get

$$\begin{vmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{vmatrix} = k(1+1+1)^3\ (1)\ (1)\ (1)\,.$$

$4(16 - 1) - 1(4 - 1) + 1(1 - 4) = 27k$

$60 - 3 - 3 = 27\,k \Rightarrow k = 2.$

$|A| = 2pqr\,(p + q + r)^3$.

Example 7.26

In a triangle ABC, if $\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin A(1+\sin A) & \sin B(1+\sin B) & \sin C(1+\sin C) \end{vmatrix} = 0,$

prove that ΔABC is an isosceles triangle.

Solution :

By putting $\sin A = \sin B$, we get

$$\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin A & 1+\sin C \\ \sin A(1+\sin A) & \sin A(1+\sin A) & \sin C(1+\sin C) \end{vmatrix} = 0$$

That is, by putting $\sin A = \sin B$ we see that, the given equation is satisfied.

Similarly by putting $\sin B = \sin C$ and $\sin C = \sin A$, the given equation is satisfied.

Thus, we have $A = B$ or $B = C$ or $C = A$.

In all cases atleast two angles are equal. Thus the triangle is isosceles.

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